

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			All three for 1 mark A - Cerebral cortex / cerebrum / frontal lobe B - Cerebellum C - Medulla oblongata	1			1		
	(b)	(i)		MRI scan shows {structural anatomy of the brain/pictures which are static} and PET scans shows areas of the brain which are functioning at a particular time (1) MRI wouldn't show active areas of the brain, as language is an active process (1)	2			2		
		(ii)		Any 5 x (1) from: A. {Broca's area is the same in both / same colour on scan} as both produce language (1) B. Increased use of motor cortex in BSL / more activity in that area in scan (1) C. because BSL involves more movement than speech alone (1) D. Increased {use/ activity} of the <u>occipital</u> lobe for {vision in BSL / visual stimuli} (1) E. Decreased use of the auditory cortex in BSL/ less activity in that area(1) F. as no link to sound and ear/ no sound detected (1)		2	3	5		
		(iii)		Any 2 x (1) from: Age when became deaf (1) Reason for deafness/ or description of (1) Level of deafness (1)		2		2		2

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		(iv)		Deaf people may exhibit neuroplasticity /'brain has adapted to form new connections as there is no auditory stimuli being received (1) Auditory cortex involved in interpretation of BSL (as near to Wernicke's area) (1) New connections/different connections to hearing between auditory cortex and {Broca's / Wernicke's} area (1)	1		2	3		
	(c)	(i)		Building a nest / to attract a mate/increase reproductive success / protect offspring/eggs / reduce competition with other males for mating (1)	1			1		
		(ii)		sign stimulus - a stimulus which elicits/causes a FAP (fixed action pattern) in the sticklebacks/ produced by one individual, causes a response in a second individual (1) Red belly triggers the aggressive behaviour/ attack/ bite (1) Data reference twice as many bites seen when red bellied model used/ more aggression seen when red bellied model used (1)	1	2		3		
		(iii)		5.6 = 2 marks 5.59 = 1 mark $\sqrt{\frac{343.3}{11}} = 1 \text{ mark}$		2		2	2	
		(iv)		Reduce confidence in conclusion as SDs overlap (1)		1		1		1
				Question 9 total	6	9	5	20	2	3